JOEL LAL

Computer Science & Artificial Intelligence Undergraduate | Desktop Software & AI Prototyping

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PROFESSIONAL SUMMARY

Third-year CS & AI undergraduate at Murdoch University Dubai experienced in multithreaded desktop GUIs, acoustic fingerprinting engines, and multi-agent AI systems. Co-author of research presented at ITT 2026 (HCT Dubai); seeking software developer internships.

EDUCATION

Murdoch University Dubai | *BSc in Computer Science & Artificial Intelligence* — **Coursework: April 2027** *(Dubai, UAE)*

- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Database Systems, Machine Learning Foundations, OOP. Coursework completion: April 2027; conferral mid-2027.

GEMS Millennium School | *Senior Secondary (CBSE) | Computer Science & Multimedia* — **Completed: 2023** *(Sharjah, UAE)*

RESEARCH & CONFERENCE PUBLICATIONS

Adaptive Multi-Agent Framework for Personalised Learning Support | *Co-Author & Presenter* — **Presented: Oct 2026** *(HCT Dubai)*

- Co-authored and presented peer-reviewed research at ITT 2026 (HCT Dubai); engineered a deterministic rule engine coordinating specialized LLMs for adaptive learning support, saving 735.6 tokens per interaction.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, Assembly, JavaScript, SQL, MicroPython, HTML/CSS.
- **Frameworks & GUI:** CustomTkinter, TkinterDnD2, Pillow, Chromaprint (fpcalc), Mutagen, OpenCV, PyTorch.
- **Systems & Tools:** Multiprocessing Pool, Worker Threading, Inno Setup, PyInstaller, Git, SQLite, Linux/CLI, REST APIs.

FEATURED TECHNICAL PROJECTS

FXsonic – Desktop Audio Tagging & Fingerprinting Suite *| Python 3.13, Chromaprint, AcoustID, SQLite | FXcodez* — **2025 – Present** *(github.com/FXcodez/FXsonic)*

- Built a standalone Windows GUI app (.exe/installer) for multi-format audio tagging; uses Chromaprint (fpcalc) acoustic fingerprinting to query AcoustID/MusicBrainz for automated ID3v2.4 tags, artwork embedding, and SQLite caching.

FXbatch – Multiprocessing Image Optimization & Icon Converter | *Python, CustomTkinter, Multiprocessing, Pillow | FXcodez* — **2025 – Present** *(github.com/FXcodez/FXbatch)*

- Built a multithreaded CustomTkinter desktop app distributing batch image resizing, AVIF/format conversion, and Windows icon (.ico) generation across CPU cores using Python's multiprocessing pool with in-memory BytesIO previews (~48 img/s).

→ Full codebases, standalone releases & projects (Hearth AI, Breachpoint, HandFlow) on e-portfolio via QR.

PROFESSIONAL & OPERATIONAL EXPERIENCE

Technical Logistics & Operations Assistant | *FX Designz* — **2023 – Present** *(Dubai / Sharjah, UAE)*

- Directing on-site AV systems engineering, signal routing, and media playback; resolving live hardware/software faults under strict event schedules while interfacing with corporate clients on staging logistics.
- Assisting on-line webinars with live stream control and simultaneous multi-client hosting assist.

LANGUAGES

English (Fluent), Malayalam (Native / Fluent), Hindi (Fluent).